

KEY

MATHEMATICS Introduction to Modern Algebra I Quiz 3.

5 pts

1) Prove that there are infinitely many primes.

Suppose not. Let $\{p_1, \dots, p_s\}$ be all primes.

Let $N = p_1 \cdot \dots \cdot p_s + 1$, then $(N, p_i) = 1$ for $1 \leq i \leq s$.

$N = p_1^{a_1} \cdot \dots \cdot p_s^{a_s}$ by prime factorization theorem. This contradicts the fact that $(N, p_i) = 1$! So there are infinitely many primes.

7 pts

2) Use the Well ordering principle to show that the GCD exists. (Recall that if $a, b \in \mathbb{Z}$ are not both 0, then the GCD $c = (a, b)$ is the unique positive number dividing a and b such that if d divides a and b then d divides c . We may write $c = ma + nb$.)

pt: Let $V = \{ma + nb \mid m, n \in \mathbb{Z}, ma + nb > 0\}$

Let $c = \min\{v \in V\}$. $c > 0$ ✓

Suppose $d \mid a$ & $d \mid b$, then clearly $d \mid ma + nb = c$ ✓

Write $c = aq + r$, $0 \leq r < a$ then $r = c - aq = d(m_0 - q) + nb$

As $r < c$, r is not in V , so $r = 0$, so $c = aq$

so $c \mid a$. Similarly one sees that $c \mid b$.